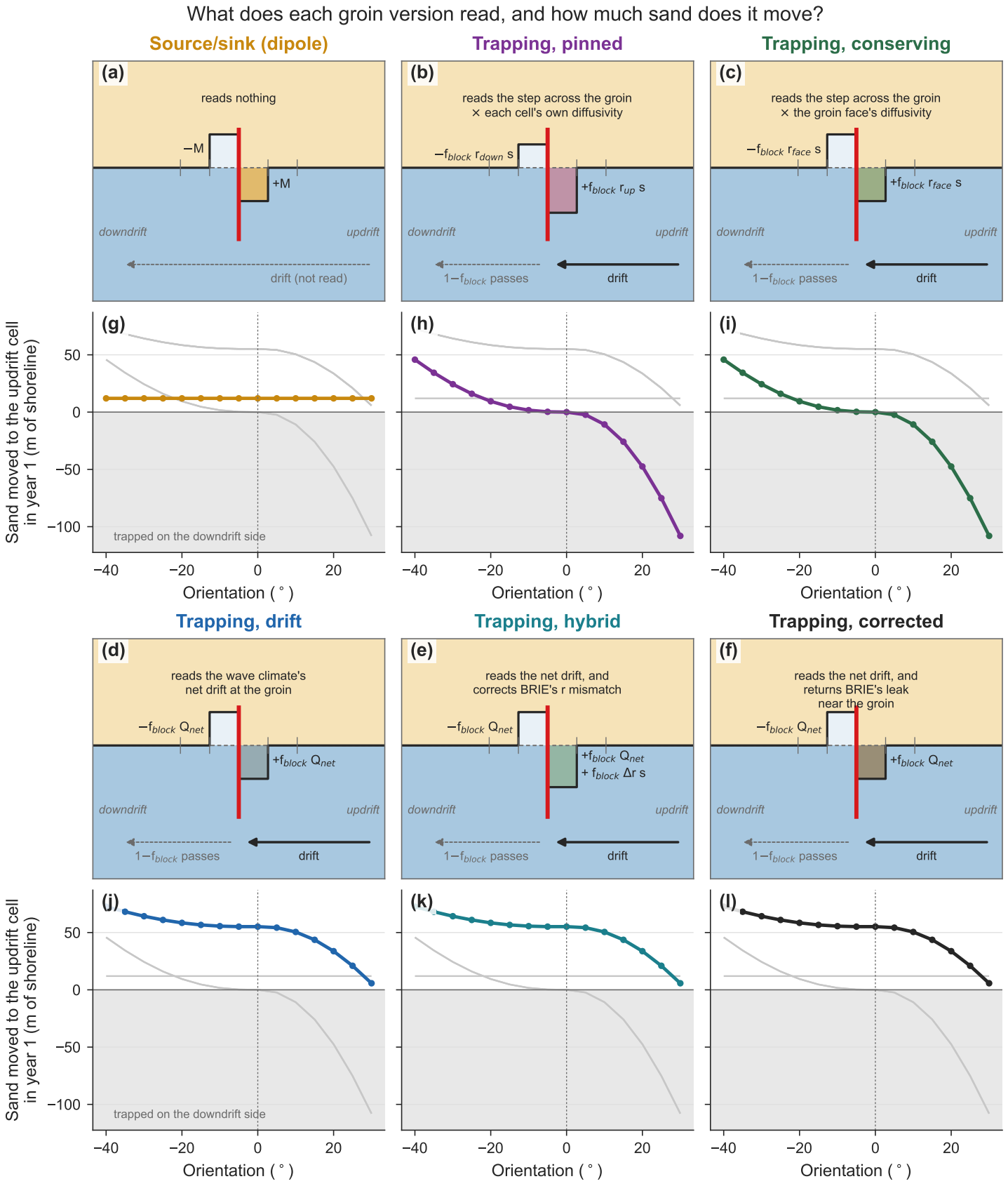
What does each groin version read, and how much sand does it move?
Source/sink (dipole)
Trapping, pinned
Trapping, conserving
(a)
reads nothing
−M
+M
downdrift
updrift
drift (not read)
(b)
reads the step across the groin
× each cell's own diffusivity
−f_block r_down s
+f_block r_up s
downdrift
updrift
1−f_block passes
drift
(c)
reads the step across the groin
× the groin face's diffusivity
−f_block r_face s
+f_block r_face s
downdrift
updrift
1−f_block passes
drift
Sand moved to the updrift cell in year 1 (m of shoreline)
(g)
(h)
(i)
trapped on the downdrift side
50
0
−50
−100
−40
−20
0
20
Orientation (°)
Trapping, drift
Trapping, hybrid
Trapping, corrected
(d)
reads the wave climate's net drift at the groin
−f_block Q_net
+f_block Q_net
downdrift
updrift
1−f_block passes
drift
(e)
reads the net drift, and corrects BRIE's r mismatch
−f_block Q_net
+f_block Q_net
+ f_block Δr s
downdrift
updrift
1−f_block passes
drift
(f)
reads the net drift, and returns BRIE's leak near the groin
−f_block Q_net
+f_block Q_net
downdrift
updrift
1−f_block passes
drift
Sand moved to the updrift cell in year 1 (m of shoreline)
(j)
(k)
(l)
trapped on the downdrift side
50
0
−50
−100
−40
−20
0
20
Orientation (°)